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SIT225: Data Capture Technologies

Activity 3.1: Arduino IoT Cloud and Dashboard with Arduino Nano 33 IoT devices

Arduino Cloud is your next exciting journey to build, control and monitor your connected projects. You can connect anything to Arduino Cloud including a wide range of compatible Arduino boards such as Arduino Nano 33 IoT or a third-party device that speaks Python. Arduino Cloud is an all-in-one IoT solution that empowers makers to create from anywhere, control their devices with stunning dashboards.

In this activity, you will connect your Arduino board to the cloud as a device, register a thing (in terms of a cloud variable) with your device, create a dashboard with a graphical widget which show value that is sent from the sketch running in your Arduino board.

Hardware Required

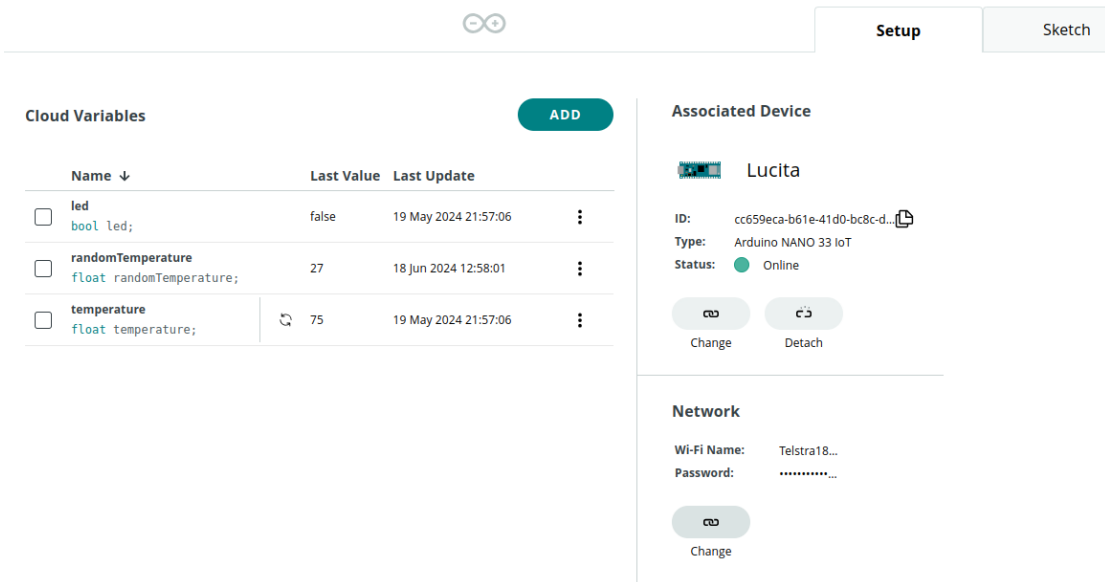
- Arduino Nano 33 IoT Board
- Wi-Fi hotspot (preferably your smartphone hotspot)
- USB cable

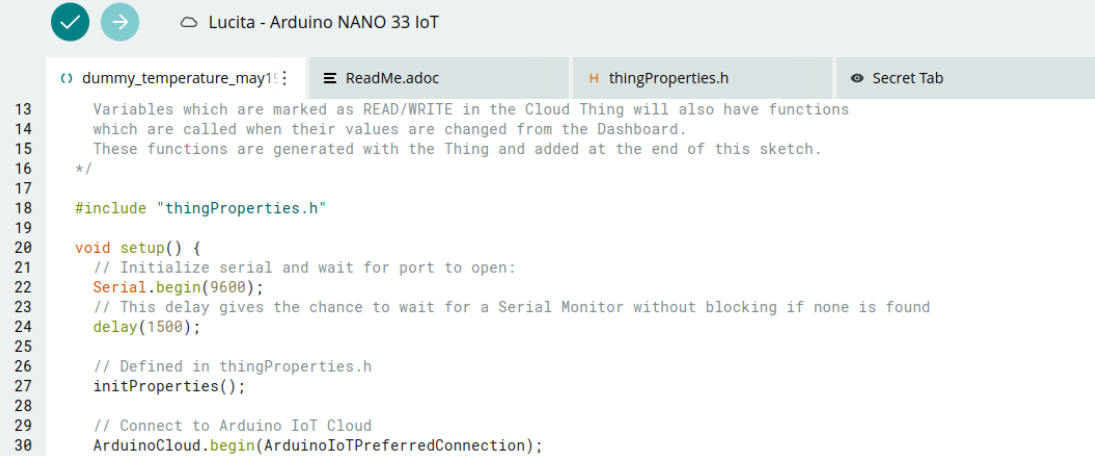
Software Required

- Arduino programming environment (Arduino IDE)
- Arduino IoT Cloud (<https://app.arduino.cc>)

Steps

Step	Action
1	Account creation: Create an account in Arduino IoT Cloud (https://app.arduino.cc). Once done, login to your account.

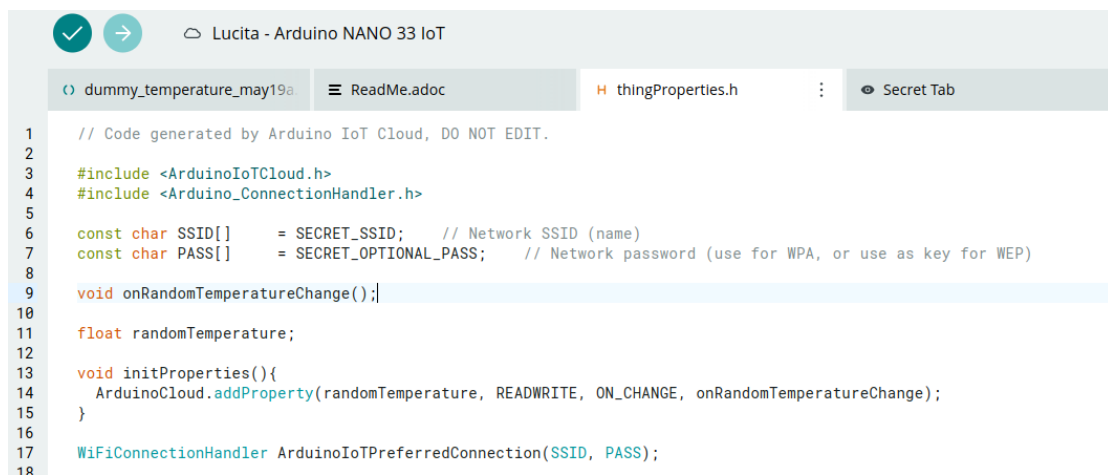
	You can follow Arduino tutorial (https://support.arduino.cc/hc/en-us/articles/360016495559-Add-and-connect-a-device-to-Arduino-Cloud) for detail. This tutorial is referred to in the steps below.
2	Add the device: Follow step 1 in the tutorial to add your Arduino Nano 33 IoT device.
3	Create and configure a Thing: Follow step 2 in the tutorial to create a Thing (sensor) and attach it to the device created in step 2 above. Note that a Thing is created as a cloud variable of specific data types. In your sketch, there should exist the same variable name and type.  The screenshot shows the Arduino Cloud interface. At the top, there are tabs for 'Setup' and 'Sketch'. Below the tabs, there are two main sections: 'Cloud Variables' and 'Associated Device'. The 'Cloud Variables' section has an 'ADD' button and a table with columns 'Name', 'Last Value', and 'Last Update'. The table lists three variables: 'led' (bool), 'randomTemperature' (float), and 'temperature' (float). The 'Associated Device' section shows the device 'Lucita' with its ID, type (Arduino NANO 33 IoT), and status (Online). There are buttons for 'Change' and 'Detach'. Below this, the 'Network' section shows the Wi-Fi Name 'Telstra18...' and Password '*****', with a 'Change' button. Consider only a single cloud variable ‘randomTemperature’ for this activity and ignore the rest (led or temperature). To keep things simple, the “Sketch” tab in the above screenshot prepares
4	Configure your Wi-Fi: In the above screenshot, the right column shows Associated Device and Network sections. You can see “Telstra” network is shown at the time this document was prepared. This should be your smartphone hotspot which is the most convenient considering the mobility – you can carry your laptop and projects without changing in the sketch.
5	Sketch tab: There is a Sketch tab at the top right corner of the screenshot above which gives you code ready to deploy in your Arduino board.



```
13 Variables which are marked as READ/WRITE in the Cloud Thing will also have functions
14 which are called when their values are changed from the Dashboard.
15 These functions are generated with the Thing and added at the end of this sketch.
16 */
17
18 #include "thingProperties.h"
19
20 void setup() {
21   // Initialize serial and wait for port to open:
22   Serial.begin(9600);
23   // This delay gives the chance to wait for a Serial Monitor without blocking if none is found
24   delay(1500);
25
26   // Defined in thingProperties.h
27   initProperties();
28
29   // Connect to Arduino IoT Cloud
30   ArduinoCloud.begin(ArduinoIoTPreferredConnection);
```

There are 3 files in the project. The first tab shows the sketch (.ino) is generally the project name. Two other sketches are header files (.h) - thingProperties.h and arduino_secrets.h.

The thingProperties.h header looks like below.



```
1 // Code generated by Arduino IoT Cloud, DO NOT EDIT.
2
3 #include <ArduinoIoTCloud.h>
4 #include <Arduino_ConnectionHandler.h>
5
6 const char SSID[] = SECRET_SSID; // Network SSID (name)
7 const char PASS[] = SECRET_OPTIONAL_PASS; // Network password (use for WPA, or use as key for WEP)
8
9 void onRandomTemperatureChange();
10
11 float randomTemperature;
12
13 void initProperties(){
14   ArduinoCloud.addProperty(randomTemperature, READWRITE, ON_CHANGE, onRandomTemperatureChange);
15 }
16
17 WiFiConnectionHandler ArduinoIoTPreferredConnection(SSID, PASS);
18
```

It shows several items including -

- Your wifi variables (such as ssid and password which you define in arduino_secrets.h header).
- The cloud variable exactly with the same name and type you have created while creating your Thing in step 3 above.
- An initialisation function called *void initProperties() {...}* which registers the Thing variable to the cloud.
- A template function *ArduinoIoTPreferredConnection(SSID, PASS)* which works behind the scenes to connect to the Arduino Cloud using the Wi-Fi you have configured.

The arduino_secrets.h header file contains Wi-Fi ssid and password. The Secret Tab shows fields where you can input Wi-Fi information. If you want to deploy using Cloud IDE shown above (called OTA upload), it needs to upgrade your

account. Instead, you can copy the sketch to your Arduino IDE installed on your computer.

```
sketch_iot_cloud.ino  arduino_secrets.h  thingProperties.h
1  #define SECRET_SSID " ";
2  #define SECRET_OPTIONAL_PASS " "
```

You can prepare the sketch and 2 header files as mentioned. Alternatively, you can download the code from here (https://github.com/deakin-deep-dreamer/sit225/tree/main/week_3).

The sketch in the GitHub link above shows the sketch below (*sketch_iot_cloud.ino*).

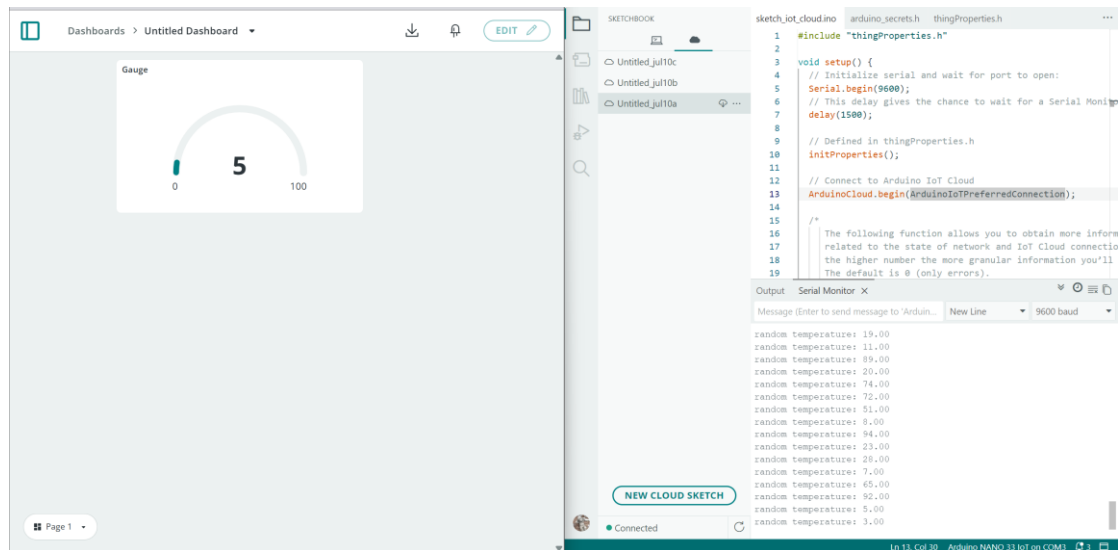
```
sketch_iot_cloud.ino  arduino_secrets.h  thingProperties.h
1  #include "thingProperties.h"
2
3  void setup() {
4      // Initialize serial and wait for port to open:
5      Serial.begin(9600);
6      // This delay gives the chance to wait for a Serial Monitor without
7      delay(1500);
8
9      // Defined in thingProperties.h
10     initProperties();
11
12     // Connect to Arduino IoT Cloud
13     ArduinoCloud.begin(ArduinoIoTPreferredConnection);
14
15     /*
16      The following function allows you to obtain more information
17      related to the state of network and IoT Cloud connection and errors
18      the higher number the more granular information you'll get.
19      The default is 0 (only errors).
20      Maximum is 4
21     */
22     setDebugMessageLevel(2);
23     ArduinoCloud.printDebugInfo();
24 }
25
26 void loop() {
27     ArduinoCloud.update();
28     // Your code here
29     randomTemperature = 1.0 * random(1, 100);
30     Serial.println("random temperature: " + String(randomTemperature));
31     delay(5*1000);
32 }
33
34 /*
35  Since Temperature is READ_WRITE variable, onRandomTemperatureChange()
36  executed every time a new value is received from IoT Cloud.
37  */
38 void onRandomTemperatureChange() {
39     // Add your code here to act upon Temperature change
40     Serial.println("--onRandomTemperatureChange");
41 }
```

	The loop function generates a random value between 1 and 100 and assigns to <i>randomTemperature</i> variable which is a cloud variable and write in serial port and waits for 5 seconds. Additional library to install: Arduino_CloudUtils <u>More info:</u> https://forum.arduino.cc/t/arduinoiotcloud-2-5-0-causes-compile-error/1373943/2
6	Deploy code to board: You can upload code to Arduino board from Arduino IDE. You can observe the random temperature readings in the serial monitor. Question: Screenshot the output of the serial monitor with random temperature values along with the initial Wi-Fi connection codes. Comment on the output lines. Answer: as we can see Connected to Internet then Connected to Arduino IoT Cloud with Thing ID: c5ad1ea5-8aea-474d-accb-ff86a34a51b7. Which mean it connect to cloud.

	sketch_iot_cloud.ino arduino_secrets.h thingProperties.h <pre> 1 #include "thingProperties.h" 2 3 void setup() { 4 // Initialize serial and wait for port to open: 5 Serial.begin(9600); 6 // This delay gives the chance to wait for a Serial Monitor without blocking if none is found 7 delay(1500); 8 9 // Defined in thingProperties.h 10 initProperties(); 11 12 // Connect to Arduino IoT Cloud 13 ArduinoCloud.begin(ArduinoIoTPreferredConnection); 14 15 /* 16 The following function allows you to obtain more information 17 related to the state of network and IoT Cloud connection and errors 18 the higher number the more granular information you'll get. 19 The default is 0 (only errors). </pre> Output Serial Monitor X Message (Enter to send message to 'Arduino NANO 33 IoT' on 'COM3') <pre> WiFi.ping(): 20 Connected to Internet random temperature: 85.00 random temperature: 65.00 random temperature: 93.00 random temperature: 33.00 random temperature: 65.00 Connected to Arduino IoT Cloud Thing ID: c5adlea5-8aea-474d-accb-ff86a34a51b7 random temperature: 85.00 random temperature: 6.00 --onRandomTemperatureChange random temperature: 60.00 random temperature: 64.00 random temperature: 83.00 random temperature: 53.00 </pre>
7	Create a dashboard: A dashboard in Arduino Cloud can be created to visualise the sensor readings sent by the Things connected to your Arduino Nano 33 IoT board. A list of Dashboard widgets is described in this tutorial (https://docs.arduino.cc/arduino-cloud/cloud-interface/dashboard-widgets). You can create a new dashboard from the Dashboards left menu items (https://app.arduino.cc/dashboards) where there are other menu items such as Devices and Things you have seen earlier. Creating a dashboard is simply choosing a dashboard widget, such as a Gauge and link it to the Thing cloud variable you have created. Once you click the Done button, you should look the Gauge widget is updating based on the value shown in the Arduino IDE serial monitor. Question: Screenshot the output of the serial monitor with random temperature values and the dashboard output so the Gauge value can be found in the serial

monitor output.

Answer:



8

Question: If you recall, there is a function *initProperties()* in *thingProperties.h* file where there is a single line -

```
13 void initProperties(){
14   ArduinoCloud.addProperty(randomTemperature, READWRITE, ON_CHANGE, onRandomTemperatureChange);
15 }
```

A function *onRandomTemperatureChange()* exists (in *sketch_iot_cloud.ino* file) to respond to *ON_CHANGE* event. Can you explore what is the use of this function and when the *ON_CHANGE* event will trigger?

Answer: for *initProperties()* it doing initing the variable *randomTemperature* to add to this code so it have variable to use. And *ON_CHANGE* trigger whenever a change happens. When web and board number don't match, it will trigger to update. After trigger *onRandomTemperatureChange()* will run.

Activity 3.2: Arduino IoT Cloud with custom Python devices

You can connect anything to Arduino Cloud including a wide range of compatible Arduino boards such as Arduino Nano 33 IoT or a third-party device that **speaks Python**.

In this activity, you will connect a custom Python board to Arduino IoT Cloud and synchronise to another cloud variable, the one you created earlier, *randomTemperature*. This will enable you to receive data from Arduino board in your Python script.

Hardware Required

Arduino Nano 33 IoT Board

Wi-Fi hotspot (preferably your smartphone hotspot)

USB cable

Software Required

Arduino programming environment (Arduino IDE)

Arduino IoT Cloud (<https://app.arduino.cc>)

Python 3

Steps

Step	Action
1	Thing & Device Configuration: Following the tutorial (https://docs.arduino.cc/arduino-cloud/guides/python), - Create a new Thing, by clicking on the "Create Thing" button. - Click on the "Select Device" in the "Associated Devices" section of your Thing. - Click on "Set Up New Device" and select the bottom category ("Manual Device"). Click continue in the next window and choose a name for your device. - Finally, you will see a new Device ID and a Secret Key generate. You can download them as a PDF. Make sure to save it as you cannot access your Secret Key again.
2	Create Variables: Follow the same tutorial mentioned above and do the following steps: While in Thing configuration, click on "Add Variable" which will open a

	new window. b. Name your variable temperature and select it to be of a float type. c. Just below the variable name, there is a link '<i>Sync with other Things</i>'. Click the link and it will show a list of all the variables you have created so far in your Arduino Cloud account in all the devices. d. Select 'randomTemperature' variable from Arduino board and click the button 'Synchronise Variables'. e. At this point the randomTemperature data sent from your Arduino Nano board will arrive in Arduino Cloud, then it will be written to Python device variable temperature. This value assignment triggers on_write event in the listening client and corresponding callback function is called.
3	Create Python script: Now you need to create a Python script to register to Arduino Cloud, register for a cloud variable called temperature and write a callback function for on_write event handling. You can write the code as below or download the code from here (https://github.com/deakin-deep-dreamer/sit225/blob/main/week_3/arduino_variable_sync.py).

```

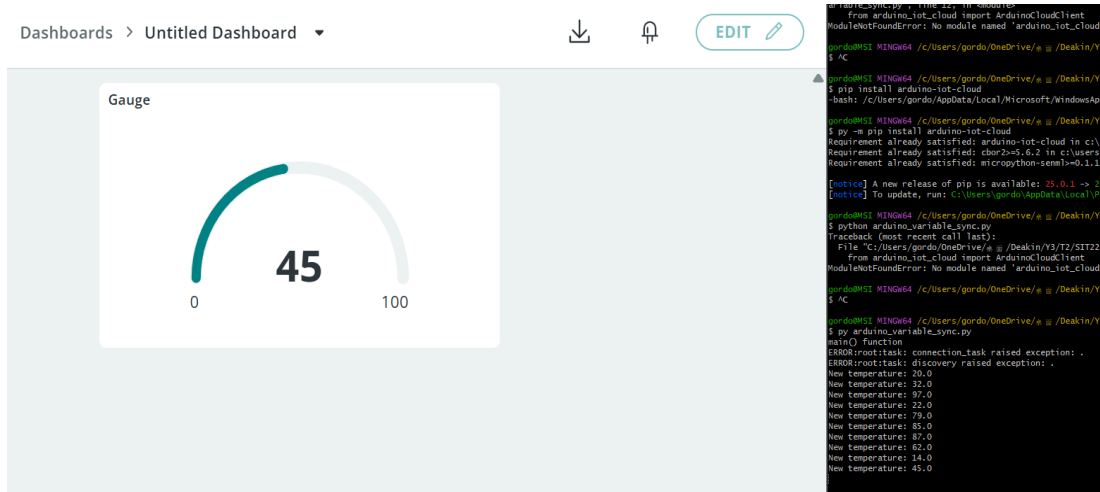
1  """
2      Requirement: arduino_iot_cloud
3      Install: pip install arduino-iot-cloud
4
5      @Ahsan Habib
6      School of IT, Deakin University, Australia.
7  """
8
9  import sys
10 import traceback
11 import random
12 from arduino_iot_cloud import ArduinoCloudClient
13 import asyncio
14
15 DEVICE_ID = "bc5c0fe9-e6ef-4eb0-90de-05032ffd9a83"
16 SECRET_KEY = "3oJYfrkmSNjM4YwKGJgV0bbBn"
17
18
19 # Callback function on temperature change event.
20 #
21 def on_temperature_changed(client, value):
22     print(f"New temperature: {value}")
23
24
25 def main():
26     print("main() function")
27
28     # Instantiate Arduino cloud client
29     client = ArduinoCloudClient(
30         device_id=DEVICE_ID, username=DEVICE_ID, password=SECRET_KEY
31     )
32
33     # Register with 'temperature' cloud variable
34     # and listen on its value changes in 'on_temperature_changed'
35     # callback function.
36     #
37     client.register(
38         "temperature", value=None,
39         on_write=on_temperature_changed)
40
41     # start cloud client
42     client.start()
43
44
45 if __name__ == "__main__":
46     try:
47         main() # main function which runs in an internal infinite loop
48     except:
49         exc_type, exc_value, exc_traceback = sys.exc_info()
50         traceback.print_tb(exc_traceback, file=print)

```

You should replace DEVICE_ID and SECRET_KEY as you were given in step 1-d above.

Question: Study the code and describe in your word how the statements match to the purpose mentioned in this step-3 above?

Answer: 1. first it setup the device by using device ID and secret key.
 2. client register temperature mean every time temperature value updates in the cloud python script gets notified instantly.
 3. then print new temp.
 4. client start mean keep running.
 So this py code is to monitor the valuable temperature in the cloud. Therefore it match what they mention in step 2.

4	Run Python script: Run the python script from command line below - <pre>\$ python arduino_variable_sync.py</pre> At this point, the command line output should show connecting to Arduino cloud and print new temperature values periodically. Question: Screenshot the Arduino Cloud Dashboard gauge showing the Arduino side randomTemperature value and screenshot the Python command-line output showing similar values. Note that there might be some lag due to network connectivity. Explain your answer accordingly. Answer:  The screenshot shows the Arduino Cloud Dashboard with a gauge labeled 'Gauge' displaying a value of 45. To the right, a terminal window shows the command-line output of the Python script. The output includes the installation of the 'arduino-iot-cloud' package and the execution of the 'arduino_variable_sync.py' script. The script prints the current temperature value, which is 45.0, and then continues to print new temperature values periodically. They are both 45.since py script is tracking the temperature value.
5	Question: Research how you can download data from Arduino IoT Cloud, say the <i>randomTemperature</i> variable if you continue the setup running for 10 minutes or so? Answer: I don't see any download or export button in the cloud, what we can do is use the py script to record down data for 10 mins and let it save it to a csv or other file.
6	Question: Discuss how you can save the data while you are receiving in Python script in a file? You can discuss in a group and come up with a solution. Hint: An algorithm of appending data can be as below. Write Python code for the algorithm. You can put the code in the call-back function <i>on_temperature_changed()</i>.

Algorithm: Append timestamp and data value to a file:

- a. Open a file in append mode
- b. Create a CSV string <timestamp>, <value> <NEWLINE>
- c. Call write function and pass CSV string, otherwise, call write_line function and pass the CSV string removing the ending <NEWLINE> (otherwise, a blank new line will be written to file).
- d. Call flash to push data to be written to file immediately.
- e. Close the file.

You can modify the algorithm to make it efficient, such as instead of opening/closing the file every time, move them to the beginning and end of the execution and keeping the CSV string formation and writing and flashing to file inside the callback function.

Answer: during this task I have add function to record down the data in a csv file. first I import datetime, for showing the date. In on_tempertrue_change function, I add log file (the file to write in), and first write in datetime, and write down the data into string.

In main(), I set up the logfile, and if log file new, then add header.

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Activity 4.1: Collect GPS data using mobile apps

Mobile phones have significant sensing capabilities such as GPS, acceleration and lux. In this task you will use existing free mobile apps to record some GPS data.

Hardware Required

A smartphone with GPS.

Software Required

A web browser

Either Android Geo Tracker

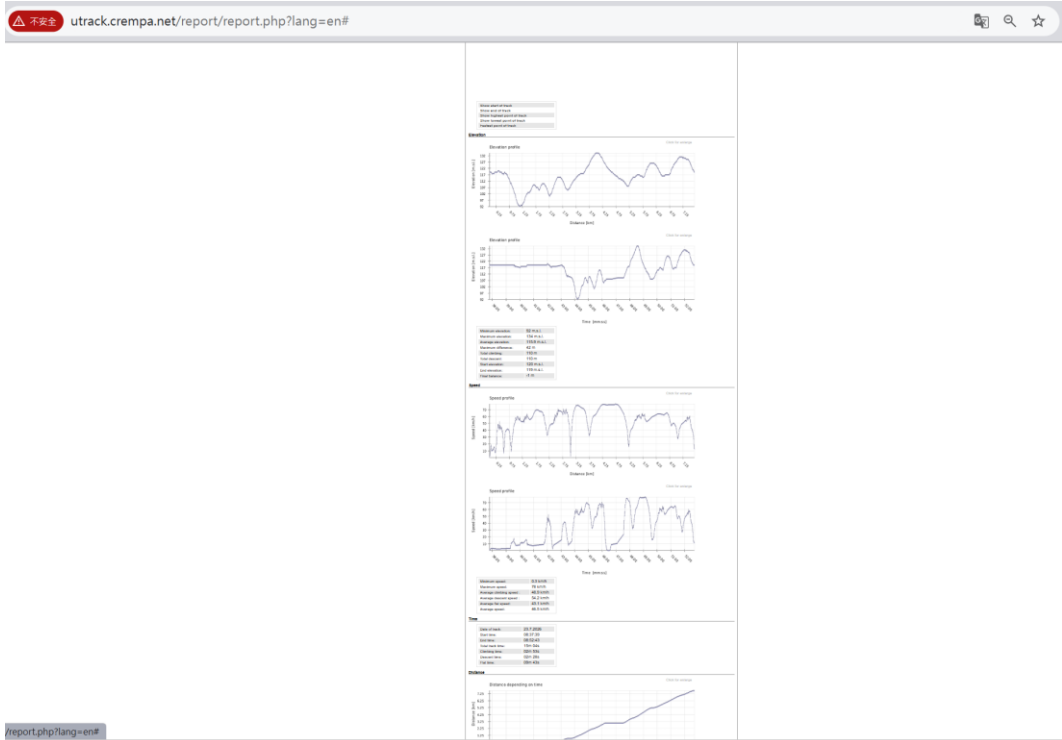
(<https://play.google.com/store/apps/details?id=com.ilyabogdanovich.geotracker&hl=en>)

Or iOS myTracks (<https://itunes.apple.com/au/app/mytracks-the-gps-logger/id358697908?mt=8>)

You may install and try out any other app well as, as long as they can track GPS & export tracks to GPX format.

Steps:

Step	Action
1	Track a journey using one of the installed mobile apps. It is best if your tracked journey spans at least 5 kilometers. Export your track to GPX format and save the file to your computer. Alternatively, you can email the .gpx file from your phone and download it to your computer Question: Upload your .gpx file to a cloud storage (such as OneDrive) and share a public link here.

	Answer: SIT225 Week4.gpx
2	Question: Open a browser on your computer and go to http://utrack.crempa.net/ . Upload your .GPX file to the site and click ‘Generate Report’. Take a screenshot and include it here. Answer:  The screenshot shows a web browser window with the URL utrack.crempa.net/report/report.php?lang=en#. The page displays a detailed report for a GPX file. It includes several line graphs for 'Elevation', 'Speed', 'Time', and 'Distance' over a period of time. Each graph has a 'Data' tab and a 'Map' tab. Below the graphs, there are summary statistics for each category. For example, the 'Elevation' section shows a maximum elevation of 100.0m and a minimum of 0.0m. The 'Speed' section shows a maximum speed of 10.0km/h. The 'Time' section shows a total time of 10.0h. The 'Distance' section shows a total distance of 10.0km. The page also includes a 'Map' tab for each graph, which shows the location of the data points on a map.
3	Question: What information can you see from the generated report? Note: Depending on the app you use to record the GPS data, timestamps could be either in local time (that is AEST if you are in Melbourne) or sometimes it could be in UTC. Answer: I used IOS myTracks, time is show in Melbourne time. In the report I can see 4 main area elevation, Speed, Time, Distance. Each area include tirp detail and a graph to see.
4	Question: Discuss in a group of 3 or 4 students to identify potential risks (more than one risk) if your data, which was collected by a service provider such as Google Map, was misused without your concern. Also suggest remedies - what kind of policies the service providers should adopt to

mitigate these risks.

Answer: if this report data leak out, people can see where I go, when did I go back home and my daily route and path walked. This could be end up me getting rob, kidnap or even murder. For my house may get invasion when I'm not at home. For law side they could use this as evidence of some crime activity that I never did. They can even do a scam call to my family or friend.

To avoid this, company should build some rule to prevent this happened like restrict company employee to read Sensity data without approve. Or host the sever in save place.

Activity 4.2: The Ethics of Data Gathering

Digital privacy is top of mind for many Australians. With weekly data breach scandals, individuals are becoming more aware and concerned about who has their data, and more importantly, who controls how that information is gathered, used and shared. In this task, you will summarise a blog post regarding what should Australian businesses know regarding the ethics of data gathering.

Steps:

Step	Action
1	Question: Study the web material which can be found here (https://www.insideinfo.com.au/business-intelligence-and-analytics/ethics-data-gathering-what-should-australian-businesses-know). Discuss data legislation in Australia and (un)ethical use of customer data. Reflect your viewpoint. Answer: This include 1. 1988 privacy act: which company need to follow APPs and if serious data leakage happened, company must report to that person. I agree this because when leakage happened company contact person let that person to know and do action to stop increase their loss in that leakage. 2. Company only collect necessarily data and tell user what data they are collecting. By doing this customer can use the service or product safely and let user decide if they want to use or not by the collect data. 3. Allow user to Anonymity and don't link personal detail to that fake account. It good for user to use if they don't want people are using.

Activity 4.3: The ethical dilemma of self-driving cars

Self-driving cars are already cruising the streets today. And while these cars will ultimately be safer and cleaner than their manual counterparts, they can't completely avoid accidents altogether. In this video, you will explore and reflect on how the car should be programmed if it encounters an unavoidable accident.

Steps:

Step	Action
1	Question: Study the web material which can be found here (https://www.youtube.com/watch?v=ixloDYVfKA0). Reflect critically the ethical considerations and discuss to what extent you agree or disagree with justification. Answer: the video is about auto car driving ethical question, I think it a good question to think about it, in extreme case, If must hit one, I will choose the way that every one is most likely to be survive. But one step backward I think there are technology that can prevent this kind of case from happening. Like set the bigger distance to the front car. Avoid car being box, decrease speed or cut len... There are no correct way to answer this question what we can do is try our best to avoid stuck in this situation.

Activity 4.4: The ethical dilemma of abusing generative language models

The use of generative language models such as ChatGPT is increasing in our daily life. A specific question can be asked in the chat-bot command prompt which generates related answers. Students often find the chat-bot a handy tool to ask any study related questions to get a direct answer and feels reluctant to search online and compile an answer by navigating through several websites. While the use of chat-bots are useful, their improper use can limit the students ability to learn. For example, using chat-bots to generate a direct answer to parts of an assignment should be avoided considering an ethical, as well as productive point-of-view.

Steps:

Step	Action
1	Question: How you feel chat-bots (such as ChatGPT) should be used in the context of solving an assignment question? Reflect critically the ethical and learning considerations. Discuss the question in a group of 3 to 4 students and share your and some of the other students' thoughts that attracted your attention. Answer: what I think is AI chat bot is good. But it a dangerous tool to use. Talking about me, for my case I can feel I am getting Dummer. Although it gives you answer which I gain knowledge, AI give us answer so fast that we don't even have time to think why is that. I would call this transferring human intelligence to knowledge. And one more thing is I am losing creativity, since everyone is using AI to generate image. All art work look the same. We should use AI as a helper. It is ok to generate the answer but give yourself some time to think why is the answer is that and fully understand the answer.

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Activity 5.1: Firebase Realtime database

The Firebase Realtime Database is a cloud-hosted NoSQL database that lets you store and sync data between your users in real-time. Data is stored as JSON and synchronized in real-time to every connected client. In this activity, you will set up and perform operations such as queries and updates on the database using Python programming language.

Hardware Required

No hardware is required.

Software Required

Firebase Realtime database
Python 3

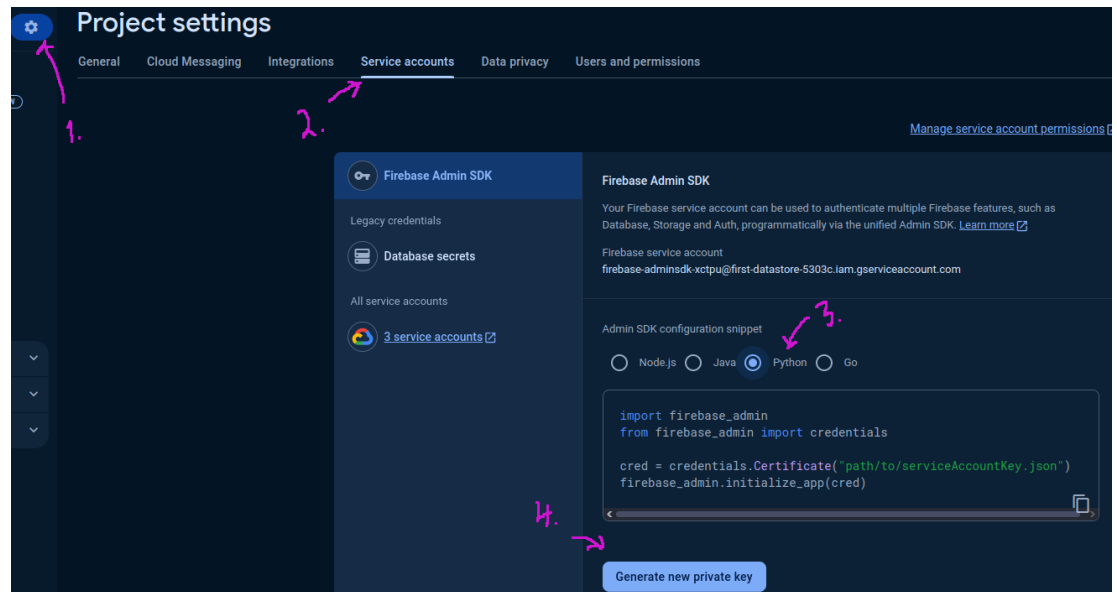
Steps

Step	Action
1	Create an Account: First, you will need to create an account in the Firebase console, follow instructions in the official Firebase document (https://firebase.google.com/docs/database/rest/start).
2	Create a Database: Follow the above Firebase document to create a database. When you click on Create Database, you have to specify the location of the database and the security rules. Two rules are available – locked mode and test mode; since we will be using the database for reading, writing, and editing, we choose test mode.
3	Setup Python library for Firebase access: We will be using Admin Database API, which is available in <i>firebase_admin</i> library. Use the below command in the command line to install. You can

follow a Firebase tutorial here (<https://www.freecodecamp.org/news/how-to-get-started-with-firebase-using-python>).

```
$ pip install firebase_admin
```

Firebase will allow access to Firebase server APIs from Google Service Accounts. To authenticate the Service Account, we require a private key in JSON format. To generate the key, go to project settings, click Generate new private key, download the file, and place it in your current folder where you will create your Python script.



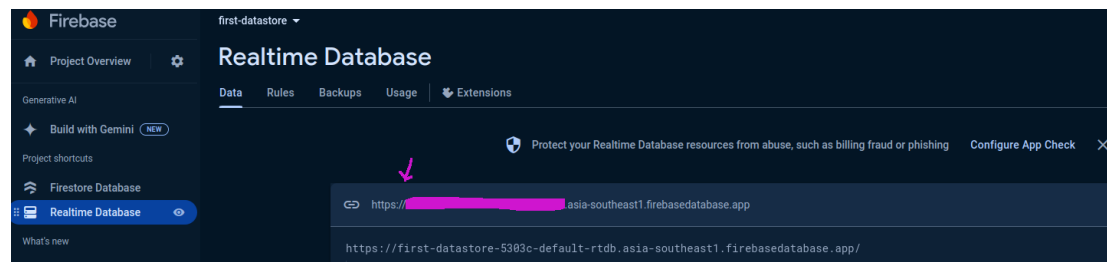
4

Connect to Firebase using Python version of Admin Database API:

A credential object needs to be created to initialise the Python library which can be done using the Python code below. Python notebook can be downloaded here (https://github.com/deakin-deep-dreamer/sit225/blob/main/week_5/firebase_explore.ipynb).

```
1 import firebase_admin
2
3 databaseURL = 'https://XXX.firebaseiodatabase.app/'
4 cred_obj = firebase_admin.credentials.Certificate(
5 | 'first-datastore-5303c-firebase-adminsdk-xctpu-c9902044ac.json'
6 | )
7 default_app = firebase_admin.initialize_app(cred_obj, {
8 | 'databaseURL': databaseURL
9 | })
```

The databaseURL is a web address to reach your Firebase database that you have created in step 2. This URL can be found in the Data tab of Realtime Database.



If you compile the code snippet above, it should do with no error.

5

Write to database Using the set() Function:

We set the reference to the root of the database (or we could also set it to a key value or child key value). Data needs to be in JSON format as below.

```

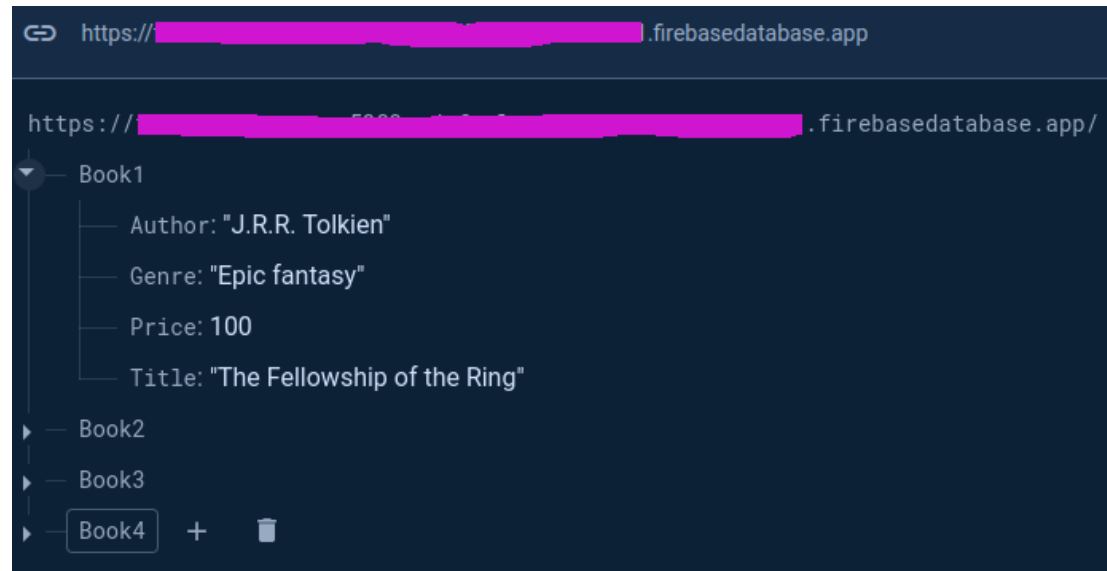
1  from firebase_admin import db
2
3  # A reference point is always needed to be set
4  # before any operation is carried out on a database.
5  #
6  ref = db.reference("/")
7
8  # JSON format data (key/value pair)
9  data = { # Outer {} contains inner data structure
10     "Book1":
11         {
12             "Title": "The Fellowship of the Ring",
13             "Author": "J.R.R. Tolkien",
14             "Genre": "Epic fantasy",
15             "Price": 100
16         },
17     "Book2":
18         {
19             "Title": "The Two Towers",
20             "Author": "J.R.R. Tolkien",
21             "Genre": "Epic fantasy",
22             "Price": 100
23         },
24     "Book3":
25         {
26             "Title": "The Return of the King",
27             "Author": "J.R.R. Tolkien",
28             "Genre": "Epic fantasy",
29             "Price": 100
30         },
31     "Book4":
32         {
33             "Title": "Brida",
34             "Author": "Paulo Coelho",
35             "Genre": "Fiction",
36             "Price": 100
37         }
38 }
39
40 # JSON format data is set (overwritten) to the reference
41 # point set at /, which is the root node.
42 #
43 ref.set(data)

```

A reference point always needed to be set where the data read/write will take place. In the code above, the reference point is set at the root of the NoSQL Document, where consider the database is a JSON tree and / is the root node

of the tree). The `set()` function writes (overwrites) data at the set reference point.

You can visualise the data in the Firebase console as below -



6 Read data using `get()` function:

Data can be read using `get()` function on the reference set beforehand, as shown below.

```

1  ref = db.reference("/") # set ref point
2
3  # query all data under the ref
4  books = ref.get()
5  print(books)
6  print(type(books))
7
8  # print each item separately
9  for key, value in books.items():
10 |     print(f"{key}: {value}")
11
12
13 # Query /Book1
14 ref = db.reference("/Book1")
15 books = ref.get()
16 print(books)

```

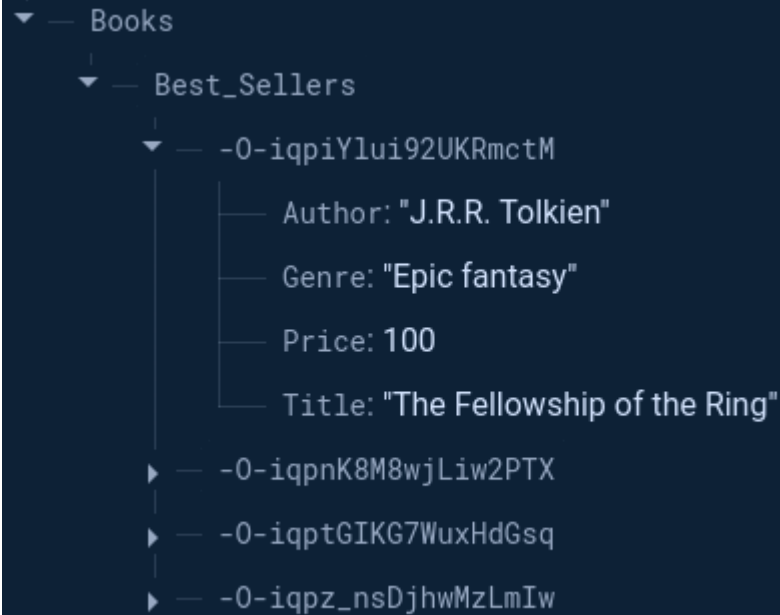
✓ 0.3s

```

{'Book1': {'Author': 'J.R.R. Tolkien', 'Genre': 'Epic fantasy', 'Price': 100, 'Title': 'The Fellowship of the Ring'},
'Book2': {'Author': 'J.R.R. Tolkien', 'Genre': 'Epic fantasy', 'Price': 100, 'Title': 'The Hobbit'},
'Book3': {'Author': 'J.R.R. Tolkien', 'Genre': 'Epic fantasy', 'Price': 100, 'Title': 'The Lord of the Rings'},
'Book4': {'Author': 'Paulo Coelho', 'Genre': 'Fiction', 'Price': 100, 'Title': 'Brida'}

```

	Consider the reference set in line 1 and the output compared to the reference set at line 14 and the bottom output line to understand the use of <code>db.reference()</code> and <code>ref.get()</code> .
7	Write to database Using the push() Function: The <code>push()</code> function saves data under a <i>unique system generated key</i>. This is different than <code>set()</code> where you set the keys such as Book1, Book2, Book3 and Book4 under which the content (author, genre, price and title) appears. Let's try to push the same data in the root reference. Note that since we already has data under root / symbol, setting (or pushing) in the same reference point will eventually rewrite the original data. <pre>1 # Write using push() function 2 # Note that a set() is called on top of push() 3 # 4 ref = db.reference("/") 5 ref.set({ 6 "Books": 7 { 8 "Best_Sellers": -1 9 } 10 }) 11 12 ref = db.reference("/Books/Best_Sellers") 13 14 for key, value in data.items(): 15 ref.push().set(value)</pre> ✓ 2.0s The output will reset the previous data set in / node. The current data is shown below.



As you can see, under /Books/Best_Sellers there are 4 nodes where the node head (or node ID) is a randomly generated key which is due to the use of push() function. When data key does not matter, the use of push() function is desirable.

8

Update data:

Let's say the price of the books by J. R. R. Tolkien is reduced to 80 units to offer a discount. The first 3 books are written by this author, and we want to apply for a discount on all of them.

```

1  # Update data
2  #
3  # Requirement: The price of the books by
4  # J. R. R. Tolkien is reduced to 80 units to
5  # offer a discount.
6  #
7  ref = db.reference("/Books/Best_Sellers/")
8  best_sellers = ref.get()
9  print(best_sellers)
10 for key, value in best_sellers.items():
11     if(value["Author"] == "J.R.R. Tolkien"):
12         value["Price"] = 90
13         ref.child(key).update({"Price":80})
✓ 0.9s
  
```

As you can see, the author name is compared and the new price is set in the best_sellers dictionary and finally, an update() function is called on the ref, however, the current ref is a '/Books/Best_Sellers/', so we need to locate the

child under the ref node, so `ref.child(key)` is used in line 13. The output is shown below with a discounted price.



9

Delete data:

Let's delete all bestseller books with J.R.R. Tolkien as the author. You can locate the node using `db.reference()` (line 4) and then locate specific record (for loop in line 6) and calling `set()` with empty data `{}` as a parameter, such as `set({})`. The particular child under the ref needs to be located first by using `ref.child(key)`, otherwise, the ref node will be removed – **BE CAREFUL**.

```

1 # Let's delete all best seller books
2 # with J.R.R. Tolkien as the author.
3 #
4 ref = db.reference("/Books/Best_Sellers")
5
6 for key, value in best_sellers.items():
7     if(value["Author"] == "J.R.R. Tolkien"):
8         ref.child(key).set({})

```

This keeps only the other author data, as shown below.



If ref.child() not used, as shown the code below, all data will be removed.

```

1 ref = db.reference("/Books/Best_Sellers")
2 ref.set({})

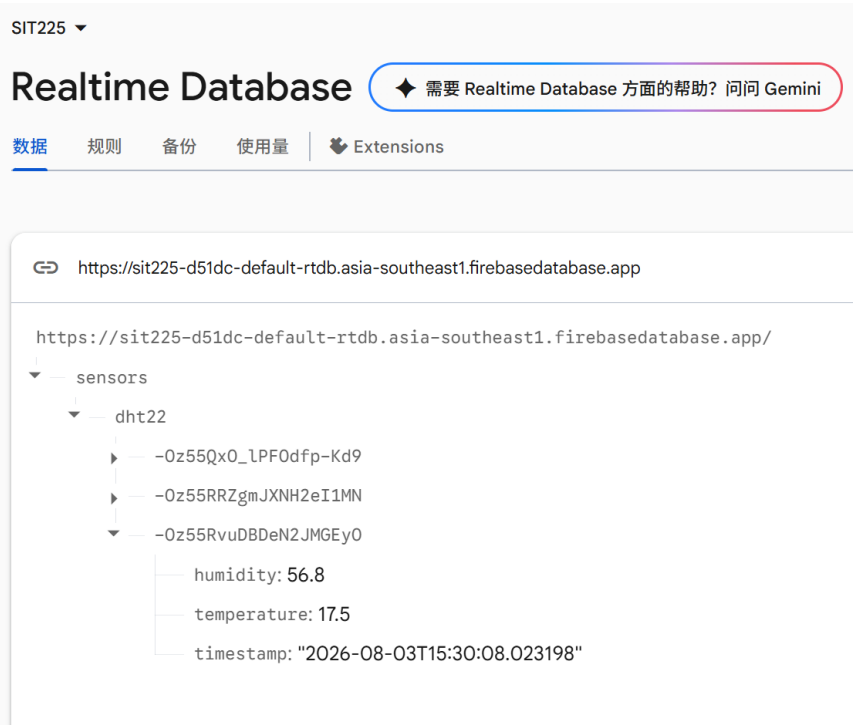
```

Now in Firebase console you will see no data exists.

10 **Question:** Run all the cells in the Notebook you have downloaded in Step 4, fill in the student information at the top cell of the Notebook. Convert the Notebook to PDF and merge with this activity sheet PDF.

Answer: Convert the Notebook to PDF and merge with this activity sheet PDF.

10 **Question:** Create a sensor data structure for DHT22 sensor which contains attributes such as sensor_name, timestamp, temperature and humidity. Remember there will be other sensors with different sensor variables such as DHT22 has 2 variables, accelerometer sensor has 3. For each such sensor, you will need to gather data over time. Discuss how you are going to handle multiple data values in JSON format? Justify your design.

	Answer: I will reuse the template code above to improve to get sensor data. For my data structure first I crate a folder called sensors to save all sensor then dht22 folder to save all record. Inside the folder it have all record using the structure “timestamp”, “temperature” and “humidity”. For adding more sensor I can just add folder in sensors.
11	Question: Generate some random data for DHT22 sensor, insert data to database, query all data and screenshot the output here. Answer:  The screenshot shows the Google Cloud Realtime Database interface for a project named SIT225. The main heading is 'Realtime Database' with a link to get help. Below the heading are tabs for '数据' (Data), '规则' (Rules), '备份' (Backups), '使用量' (Usage), and 'Extensions'. The '数据' tab is selected, showing a tree view of the database structure. The path is: sensors > dht22 > -Oz55RvuDBDeN2JMGEy0. The data for this node is: humidity: 56.8, temperature: 17.5, and timestamp: "2026-08-03T15:30:08.023198".
12	Question: Generate some random data for the SR04 Ultrasonic sensor, insert data to database, query all data and screenshot the output here. Answer:

	Realtime Database ◆ 需要 Realtime Database 方面的帮助? 问问 Gemini 数据 规则 备份 使用量 Extensions https://sit225-d51dc-default-rtdb.asia-southeast1.firebaseio.com https://sit225-d51dc-default-rtdb.asia-southeast1.firebaseio.com/ sensors dht22 sr04 -0z5C2vJgP50Kp1X2x-y -0z5C2xJMIAT03C1rGwa -0z5C3-yFnj3bjfaGnIP distance_cm: 88.94 timestamp: "2026-08-03T15:59:00.983699"
13	Question: Firebase Realtime database generates events on data operations. You can refer to section 'Handling Realtime Database events' in the document (https://firebase.google.com/docs/functions/database-events?gen=2nd). Discuss in the active learning session and summarise the idea of database events and how it is handled using Python SDK. Note that these events are useful when your sensors (from Arduino script) store data directly to Firebase Realtime database and you would like to track data update actions from a central Python application such as a monitoring dashboard. Answer: Firebase Realtime Database is a reactive, real-time cloud database. When data operations occur such as write, update or delete, Firebase automatically triggers Database Events. What I did above is push reading like DDHT22 and SR04 sensor directly to firebase. It allows external app or device to react immediately without refreshing the database. The <code>firebase_admin</code> Python SDK handles events using Event Listeners via the <code>db.reference().listen(callback)</code> method.

Activity 5.2: Data wrangling

Data wrangling is the process of converting raw data into a usable form. The process includes collecting, processing, analyzing, and tidying the raw data so that it can be easily read and analyzed. In this activity, you will use the common library in python, "pandas".

Hardware Required

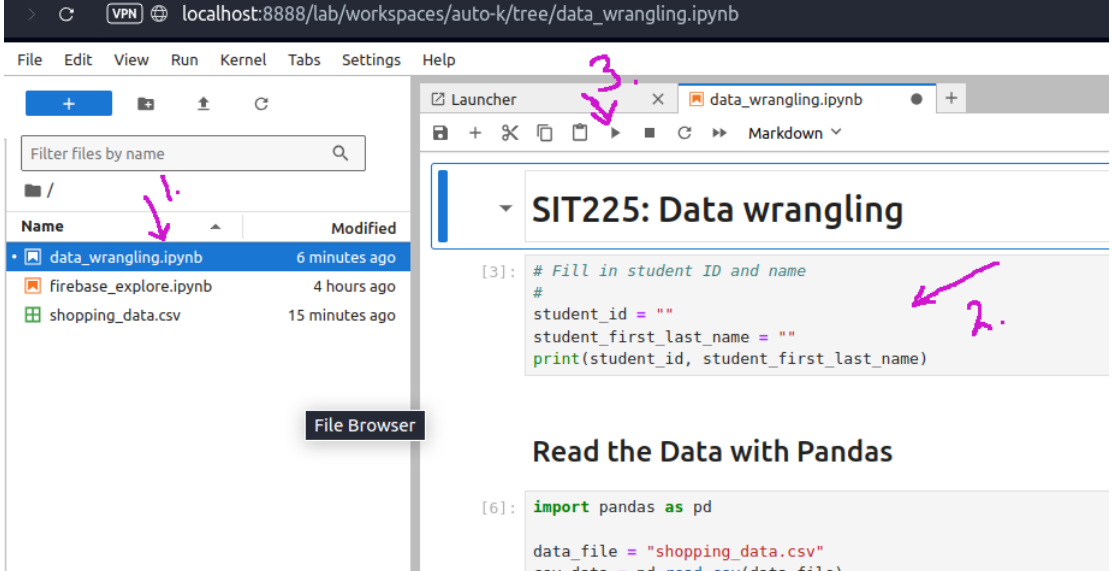
No hardware is required.

Software Required

Python 3
Pandas Python library

Steps

Step	Action
1	Install Pandas using the command below. Most likely you already have Pandas installed if you have installed Python using Anaconda distribution (https://www.anaconda.com/download). <pre>\$ pip install pandas</pre> A Python notebook is shared in the GitHub link (https://github.com/deakin-deep-dreamer/sit225/tree/main/week_5). There will be a data_wrangling.ipynb, shopping_data.csv and shopping_data_missingvalue.csv files among others. Download the week_5 folder in your computer, open a command prompt in that folder, and write the command below in the command line: <pre>\$ jupyter lab</pre> This will open Python Jupyter Notebook where in the left panel you can see the files (labeled as 1 in figure).

	 Each cell contains Python code (labeled as 2 in figure), you can run a cell by clicking on the cell, so the cursor appears in that cell and then click on the play button at the top of the panel (labeled as 3 in the figure).
2	Question: Run each cell to produce output. Follow instructions in the notebook to complete codes in some of the cells. Convert the notebook to PDF from menu File > Save and Export Notebook As > PDF. Convert this activity sheet to PDF and merge with the notebook PDF. Answer: There is no answer to write here. You have to answer in the Jupyter Notebook.
3	Question: Once you went through the cells in the Notebook, you now have a basic understanding of data wrangling. Pandas are a powerful tool and can be used for reading CSV data. Can you use Pandas in reading sensor CSV data that you generated earlier? Describe if any modification you think necessary? Answer: I think it possible to read the data we just input in firebase however I need to decide a way to get the data first like should I export it to csv file first or I can directly get from the database? After getting the data I also need to check if any Nan value and clean them. further I can change my timeframe format.
4	Question: What do you understand of the Notebook section called Handling Missing Value? Discuss in group and briefly summarise different missing value imputation methods and their applicability on different data conditions.

	Answer: if a dataset have missing value or NAN value, some function like <code>mean()</code> or math calculation won't work and return error. So we can either delete the or using so math method to fill in the gap without changing too much of overall performance. Like using <code>max()</code>, <code>min()</code>, <code>mean</code>, <code>median()</code> or <code>avg()</code>. Depend on which is most suitable for the project.
--	---

```
In [23]: # Fill in student ID and name
#
student_id = "s223532749"
student_first_last_name = "Gordon Tse"
print(student_id, student_first_last_name)
```

s223532749 Gordon Tse

```
In [ ]: # Install libraris, if not yet.
# ! pip install firebase_admin pandas

import firebase_admin

databaseURL = 'https://sit225-d51dc-default-rtdb.asia-
southeast1.firebaseio.com/'
cred_obj = firebase_admin.credentials.Certificate(
    'sit225-d51dc-firebase-adminsdk-fbsvc-bb6c4788ce.json'
)
default_app = firebase_admin.initialize_app(cred_obj, {
    'databaseURL':databaseURL
})
```

```
-----
-----
ValueError                                Traceback (most
recent call last)
Cell In[24], line 10
      6 databaseURL = 'https://sit225-d51dc-default-rtdb.asia-
southeast1.firebaseio.com/'
      7 cred_obj = firebase_admin.credentials.Certificate(
      8     'sit225-d51dc-firebase-adminsdk-fbsvc-
bb6c4788ce.json'
      9 )
--> 10 default_app = firebase_admin.initialize_app(cred_obj, {
      11     'databaseURL':databaseURL
      12     })

File c:\Users\gordo\anaconda3\Lib\site-
packages\firebase_admin\__init__.py:74, in
initialize_app(credential, options, name)
      71     return app
      73 if name == _DEFAULT_APP_NAME:
--> 74     raise ValueError((
      75         'The default Firebase app already exists. This
means you called '
      76         'initialize_app() more than once without
providing an app name as '
      77         'the second argument. In most cases you only
need to call '
      78         'initialize_app() once. But if you do want to
initialize multiple '
      79         'apps, pass a second argument to
initialize_app() to give each app '
      80         'a unique name.'))
      82 raise ValueError(
      83     f'Firebase app named "{name}" already exists. This
means you called '
      84     'initialize_app() more than once with the same app
name as the '
      85     'second argument. Make sure you provide a unique
name every time '

```

86 'you call initialize_app().')

ValueError: The default Firebase app already exists. This means you called initialize_app() more than once without providing an app name as the second argument. In most cases you only need to call initialize_app() once. But if you do want to initialize multiple apps, pass a second argument to initialize_app() to give each app a unique name.

```
In [25]: from firebase_admin import db

# A reference point is always needed to be set
# before any operation is carried out on a database.
#
ref = db.reference("/")

# JSON format data (key/value pair)
data = { # Outer {} contains inner data structure
    "Book1":
        {
            "Title": "The Fellowship of the Ring",
            "Author": "J.R.R. Tolkien",
            "Genre": "Epic fantasy",
            "Price": 100
        },
    "Book2":
        {
            "Title": "The Two Towers",
            "Author": "J.R.R. Tolkien",
            "Genre": "Epic fantasy",
            "Price": 100
        },
    "Book3":
        {
            "Title": "The Return of the King",
            "Author": "J.R.R. Tolkien",
            "Genre": "Epic fantasy",
            "Price": 100
        },
    "Book4":
        {
            "Title": "Brida",
            "Author": "Paulo Coelho",
            "Genre": "Fiction",
            "Price": 100
        }
}

# JSON format data is set (overwritten) to the reference
# point set at /, which is the root node.
#
ref.set(data)
```

In [26]: `ref = db.reference("/") # set ref point`

```
# query all data under the ref
books = ref.get()
print(books)
print(type(books))

# print each item separately
for key, value in books.items():
    print(f"{key}: {value}")

# Query /Book1
ref = db.reference("/Book1")
books = ref.get()
print(books)
```

```
{'Book1': {'Author': 'J.R.R. Tolkien', 'Genre': 'Epic fantasy',
'Price': 100, 'Title': 'The Fellowship of the Ring'}, 'Book2':
{'Author': 'J.R.R. Tolkien', 'Genre': 'Epic fantasy', 'Price':
100, 'Title': 'The Two Towers'}, 'Book3': {'Author': 'J.R.R.
Tolkien', 'Genre': 'Epic fantasy', 'Price': 100, 'Title': 'The
Return of the King'}, 'Book4': {'Author': 'Paulo Coelho', 'Genre':
'Fiction', 'Price': 100, 'Title': 'Brida'}}
<class 'dict'>
Book1: {'Author': 'J.R.R. Tolkien', 'Genre': 'Epic fantasy',
'Price': 100, 'Title': 'The Fellowship of the Ring'}
Book2: {'Author': 'J.R.R. Tolkien', 'Genre': 'Epic fantasy',
'Price': 100, 'Title': 'The Two Towers'}
Book3: {'Author': 'J.R.R. Tolkien', 'Genre': 'Epic fantasy',
'Price': 100, 'Title': 'The Return of the King'}
Book4: {'Author': 'Paulo Coelho', 'Genre': 'Fiction', 'Price':
100, 'Title': 'Brida'}
{'Author': 'J.R.R. Tolkien', 'Genre': 'Epic fantasy', 'Price':
100, 'Title': 'The Fellowship of the Ring'}
```

In [27]: `# Write using push() function`

```
# Note that a set() is called on top of push()
#
ref = db.reference("/")
ref.set({
    "Books":
        {
            "Best_Sellers": -1
        }
})

ref = db.reference("/Books/Best_Sellers")

for key, value in data.items():
    ref.push().set(value)
```

```
In [28]: # Update data
#
# Requirement: The price of the books by
# J. R. R. Tolkien is reduced to 80 units to
# offer a discount.
#
ref = db.reference("/Books/Best_Sellers/")
best_sellers = ref.get()
print(best_sellers)
for key, value in best_sellers.items():
    if(value["Author"] == "J.R.R. Tolkien"):
        value["Price"] = 90
        ref.child(key).update({"Price":80})

{'-Oyquin0tTFaJI1rm1GK': {'Author': 'J.R.R. Tolkien', 'Genre':
'Epic fantasy', 'Price': 100, 'Title': 'The Fellowship of the
Ring'}, '-OyquirFUS24HgPGUAjn': {'Author': 'J.R.R. Tolkien',
'Genre': 'Epic fantasy', 'Price': 100, 'Title': 'The Two Towers'},
'-OyquivKF_9LBVnsEz7r': {'Author': 'J.R.R. Tolkien', 'Genre':
'Epic fantasy', 'Price': 100, 'Title': 'The Return of the King'},
'-OyquizJown7vK-kl0zy': {'Author': 'Paulo Coelho', 'Genre':
'Fiction', 'Price': 100, 'Title': 'Brida'}}
```

```
In [29]: # Let's delete all best seller books
# with J.R.R. Tolkien as the author.
#
ref = db.reference("/Books/Best_Sellers")

for key, value in best_sellers.items():
    if(value["Author"] == "J.R.R. Tolkien"):
        ref.child(key).set({})
```

```
In [30]: # Delete all best_seller data.
#
ref = db.reference("/Books/Best_Sellers/")
best_sellers = ref.get()
print(best_sellers)
print(type(best_sellers))

{'-OyquizJown7vK-kl0zy': {'Author': 'Paulo Coelho', 'Genre':
'Fiction', 'Price': 100, 'Title': 'Brida'}}
<class 'dict'>
```

```
In [31]: ref = db.reference("/Books/Best_Sellers")
ref.set({})
```

SIT225: Data wrangling

Run each cell to generate output and finally convert this notebook to PDF.

```
# Fill in student ID and name
#
student_id = "Gordon"
student_first_last_name = "Tse"
print(student_id, student_first_last_name)
```

Gordon Tse

Read the Data with Pandas

Pandas has a dedicated function `read_csv()` to read CSV files.

Just in case we have a large number of data, we can just show into only five rows with `head` function. It will show you 5 rows data automatically.

```
import pandas as pd

data_file = "shopping_data.csv"
csv_data = pd.read_csv(data_file)

print(csv_data)

# show into only five rows with head function
print(csv_data.head())
```

	CustomerID	Genre	Age	Annual Income (k\$)	Spending Score (1-100)
0	1	Male	19	15	
1	2	Male	21	15	
2	3	Female	20	16	
3	4	Female	23	16	
4	5	Female	31	17	
...	...	...	...	...	...
195	196	Female	35	120	
196	197	Female	45	126	

197	198	Male	32	126	
74					
198	199	Male	32	137	
18					
199	200	Male	30	137	
83					
[200 rows x 5 columns]					
	CustomerID	Genre	Age	Annual Income (k\$)	Spending Score (1-100)
0	1	Male	19	15	39
1	2	Male	21	15	81
2	3	Female	20	16	6
3	4	Female	23	16	77
4	5	Female	31	17	40

Access the Column

Pandas has provided function `.columns` to access the column of the data source.

```
print(csv_data.columns)

# if we want to access just one column, for example "Age"
print("Age:")
print(csv_data["Age"])

Index(['CustomerID', 'Genre', 'Age', 'Annual Income (k$)',
      'Spending Score (1-100)'],
      dtype='object')
Age:
0      19
1      21
2      20
3      23
4      31
...
195    35
196    45
197    32
198    32
199    30
Name: Age, Length: 200, dtype: int64
```

Access the Row

In addition to accessing data through columns, using pandas can also access using rows. In contrast to access through columns, the function to display data from a row is the `.iloc[i]` function where `[i]` indicates the order of the rows to be displayed where the index starts from 0.

```
# we want to know what line 5 contains
print(csv_data.iloc[5])

print()

# We can combine both of those function to show row and column we
# want.
# For the example, we want to show the value in column "Age" at the
# first row
# (remember that the row starts at 0)
#
print(csv_data["Age"].iloc[1])
```

CustomerID	6
Genre	Female
Age	22
Annual Income (k\$)	17
Spending Score (1-100)	76

Name: 5, dtype: object

21

Show Data Based on Range

After displaying a data set, what if you want to display data from rows 5 to 20 of a dataset? To anticipate this, pandas can also display data within a certain range, both ranges for rows only, only columns, and ranges for rows and columns

```
print("Shows data to 5th to less than 10th in a row:")
print(csv_data.iloc[5:10])
```

Shows data to 5th to less than 10th in a row:

	CustomerID	Genre	Age	Annual Income (k\$)	Spending Score (1-100)
5	6	Female	22	17	76
6	7	Female	35	18	6
7	8	Female	23	18	94
8	9	Male	64	19	3
9	10	Female	30	19	72

Using Numpy to Show the Statistic Information

The describe() function allows to quickly find statistical information from a dataset. Those information such as mean, median, modus, max min, even standard deviation. Don't forget to install Numpy before using describe function.

```
print(csv_data.describe(include="all"))
```

	CustomerID	Genre	Age	Annual Income (k\$)	\
count	200.000000	200	200.000000	200.000000	
unique	NaN	2	NaN	NaN	
top	NaN	Female	NaN	NaN	
freq	NaN	112	NaN	NaN	
mean	100.500000	NaN	38.850000	60.560000	
std	57.879185	NaN	13.969007	26.264721	
min	1.000000	NaN	18.000000	15.000000	
25%	50.750000	NaN	28.750000	41.500000	
50%	100.500000	NaN	36.000000	61.500000	
75%	150.250000	NaN	49.000000	78.000000	
max	200.000000	NaN	70.000000	137.000000	

	Spending Score (1-100)
count	200.000000
unique	NaN
top	NaN
freq	NaN
mean	50.200000
std	25.823522
min	1.000000
25%	34.750000
50%	50.000000
75%	73.000000
max	99.000000

Handling Missing Value

For the first step, we will figure out if there is missing value.

```
print(csv_data.isnull().values.any())
```

```
print()
```

False

We will use another data source with missing values to practice this part.

```
data_missing = pd.read_csv("shopping_data_missingvalue.csv")
```

```
print(data_missing.head())
```

```
print()
```

```
print("Missing? ", data_missing.isnull().values.any())
```

	CustomerID	Genre	Age	Annual Income (k\$)	Spending Score (1-100)
0	1	Male	19.0	15.0	39.0
1	2	Male	NaN	15.0	

81.0				
2	3	Female	20.0	NaN
6.0				
3	4	Female	23.0	16.0
77.0				
4	5	Female	31.0	17.0
NaN				

Missing? True

Ways to deal with missing values.

Follow the tutorial (<https://deepnote.com/app/rickyharyanto14-3390/Data-Wrangling-w-Python-e5d1a23e-33cf-416d-ad27-4c3f7f467442>). It includes -

1. Delete data
 - deleting rows
 - pairwise deletion
 - delete column
2. imputation
 - time series problem
 - Data without trend with seasonality (mean, median, mode, random)
 - Data with trend and without seasonality (linear interpolation)
 - general problem
 - Data categorical (Make NA as multiple imputation)
 - Data numerical or continuous (mean, median, mode, multiple imputation and linear regression)

Filling with Mean Values

The mean is used for data that has a few outliers/noise/anomalies in the distribution of the data and its contents. This value will later fill in the empty value of the dataset that has a missing value case. To fill in an empty value use the `fillna()` function

```
print(data_missing.mean())
```

```
"""
```

Question: This code will generate error. Can you explain why and how it can be solved?

Move on to the next cell to find one way it can be solved.

Answer: The code will generate an error because `mean()` function is being called on a Dataset that contains missing values. The `mean()` function cannot compute the mean of columns that have NaN values, which leads to an error.

```
"""
```

```
-----  
-----  
TypeError                                Traceback (most recent call  
last)
```

```
Cell In[12], line 1
```

```
----> 1 print(data_missing.mean())  
      3 ""  
      4  
      5 Question: This code will generate error. Can you explain why  
and how it can be solved?  
      (...)   
      9  
     10 ""
```

```
File c:\Users\gordo\anaconda3\Lib\site-packages\pandas\core\  
frame.py:11335, in DataFrame.mean(self, axis, skipna, numeric_only,  
**kwargs)
```

```
    11327 @doc(make_doc("mean", ndim=2))  
    11328 def mean(  
    11329     self,  
    (...)   
    11333     **kwargs,  
    11334 ):  
> 11335     result = super().mean(axis, skipna, numeric_only,  
**kwargs)  
    11336     if isinstance(result, Series):  
    11337         result = result.__finalize__(self, method="mean")
```

```
File c:\Users\gordo\anaconda3\Lib\site-packages\pandas\core\  
generic.py:11992, in NDFrame.mean(self, axis, skipna, numeric_only,  
**kwargs)
```

```
    11985 def mean(  
    11986     self,  
    11987     axis: Axis | None = 0,  
    (...)   
    11990     **kwargs,  
    11991 ) -> Series | float:  
> 11992     return self._stat_function(  
    11993         "mean", nanops.nanmean, axis, skipna, numeric_only,  
**kwargs  
    11994     )
```

```
File c:\Users\gordo\anaconda3\Lib\site-packages\pandas\core\  
generic.py:11949, in NDFrame._stat_function(self, name, func, axis,  
skipna, numeric_only, **kwargs)
```

```
    11945 nv.validate_func(name, (), kwargs)  
    11947 validate_bool_kwarg(skipna, "skipna", none_allowed=False)  
> 11949 return self._reduce(  
    11950     func, name=name, axis=axis, skipna=skipna,  
numeric_only=numeric_only
```

```
11951 )
```

```
File c:\Users\gordo\anaconda3\Lib\site-packages\pandas\core\
frame.py:11204, in DataFrame._reduce(self, op, name, axis, skipna,
numeric_only, filter_type, **kwargs)
```

```
11200     df = df.T
11202 # After possibly _get_data and transposing, we are now in the
11203 # simple case where we can use BlockManager.reduce
> 11204 res = df._mgr.reduce(blk_func)
11205 out = df._constructor_from_mgr(res, axes=res.axes).iloc[0]
11206 if out_dtype is not None and out.dtype != "boolean":
```

```
File c:\Users\gordo\anaconda3\Lib\site-packages\pandas\core\internals\
managers.py:1459, in BlockManager.reduce(self, func)
```

```
1457 res_blocks: list[Block] = []
1458 for blk in self.blocks:
-> 1459     nbs = blk.reduce(func)
1460     res_blocks.extend(nbs)
1462 index = Index([None]) # placeholder
```

```
File c:\Users\gordo\anaconda3\Lib\site-packages\pandas\core\internals\
blocks.py:377, in Block.reduce(self, func)
```

```
371 @final
372 def reduce(self, func) -> list[Block]:
373     # We will apply the function and reshape the result into a
single-row
374     # Block with the same mgr_locs; squeezing will be done at
a higher level
375     assert self.ndim == 2
--> 377     result = func(self.values)
379     if self.values.ndim == 1:
380         res_values = result
```

```
File c:\Users\gordo\anaconda3\Lib\site-packages\pandas\core\
frame.py:11136, in DataFrame._reduce.<locals>.blk_func(values, axis)
```

```
11134     return np.array([result])
11135 else:
> 11136     return op(values, axis=axis, skipna=skipna, **kwargs)
```

```
File c:\Users\gordo\anaconda3\Lib\site-packages\pandas\core\
nanops.py:147, in bottleneck_switch.__call__.<locals>.f(values, axis,
skipna, **kwargs)
```

```
145     result = alt(values, axis=axis, skipna=skipna, **kwargs)
146 else:
--> 147     result = alt(values, axis=axis, skipna=skipna, **kwargs)
149 return result
```

```
File c:\Users\gordo\anaconda3\Lib\site-packages\pandas\core\
nanops.py:404, in _datetimelike_compat.<locals>.new_func(values, axis,
skipna, mask, **kwargs)
```

```

    401 if datetimelike and mask is None:
    402     mask = isna(values)
--> 404 result = func(values, axis=axis, skipna=skipna, mask=mask,
**kwargs)
    406 if datetimelike:
    407     result = _wrap_results(result, orig_values.dtype,
fill_value=iNaT)

```

```

File c:\Users\gordo\anaconda3\Lib\site-packages\pandas\core\
nanops.py:720, in nanmean(values, axis, skipna, mask)
    718 count = _get_counts(values.shape, mask, axis,
dtype=dtype_count)
    719 the_sum = values.sum(axis, dtype=dtype_sum)
--> 720 the_sum = _ensure_numeric(the_sum)
    722 if axis is not None and getattr(the_sum, "ndim", False):
    723     count = cast(np.ndarray, count)

```

```

File c:\Users\gordo\anaconda3\Lib\site-packages\pandas\core\
nanops.py:1678, in _ensure_numeric(x)
    1675 inferred = lib.infer_dtype(x)
    1676 if inferred in ["string", "mixed"]:
    1677     # GH#44008, GH#36703 avoid casting e.g. strings to numeric
-> 1678     raise TypeError(f"Could not convert {x} to numeric")
    1679 try:
    1680     x = x.astype(np.complex128)

```

TypeError: Could not convert

```

['MaleMaleFemaleFemaleFemaleFemaleFemaleFemaleMaleFemaleMaleFemaleFema
leFemaleMaleMaleFemaleMaleMaleFemaleMaleMaleFemaleMaleFemaleMaleFemale
MaleFemaleFemaleMaleFemaleMaleMaleFemaleFemaleFemaleFemaleFemaleFemale
FemaleMaleMaleFemaleFemaleFemaleFemaleFemaleFemaleFemaleFemaleFemaleMaleFema
leMaleFemaleMaleFemaleMaleFemaleMaleMaleMaleFemaleFemaleMaleMaleFemale
FemaleMaleFemaleMaleFemaleFemaleFemaleMaleMaleFemaleMaleFemaleFemaleMa
leMaleMaleFemaleFemaleMaleFemaleFemaleFemaleFemaleFemaleFemaleMaleMaleFemale
FemaleMaleFemaleFemaleMaleMaleFemaleFemaleMaleMaleMaleFemaleFemaleMale
MaleMaleMaleFemaleFemaleMaleFemaleFemaleFemaleFemaleFemaleFemaleFemaleMaleFe
maleFemaleMaleFemaleFemaleMaleMaleMaleMaleMaleMaleFemaleFemaleMaleFema
leFemaleMaleMaleFemaleFemaleMaleFemaleFemaleMaleMaleMaleFemaleFemaleMa
leMaleMaleFemaleFemaleFemaleFemaleMaleFemaleMaleFemaleFemaleFemaleMale
FemaleMaleFemaleMaleFemaleFemaleMaleMaleMaleMaleMaleFemaleFemaleMaleMa
leMaleMaleFemaleFemaleMaleFemaleFemaleMaleFemaleMaleFemaleFemaleFemale
FemaleMaleFemaleFemaleFemaleFemaleFemaleMaleMaleMale'] to numeric

```

Genre column contains string values and numerical operation mean fails.

Lets drop Genre column since for numerical calculation.

#

```

data_missing_wo_genre = data_missing.drop(columns=['Genre'])
print(data_missing_wo_genre.head())

```

	CustomerID	Age	Annual Income (k\$)	Spending Score (1-100)
0	1	19.0	15.0	39.0
1	2	NaN	15.0	81.0
2	3	20.0	NaN	6.0
3	4	23.0	16.0	77.0
4	5	31.0	17.0	NaN

```
print(data_missing_wo_genre.mean())
```

```
CustomerID      100.500000
Age             38.939698
Annual Income (k$)  61.005051
Spending Score (1-100)  50.489899
dtype: float64
```

```
print("Dataset with empty values! :")
```

```
print(data_missing_wo_genre.head(10))
```

```
data_filling=data_missing_wo_genre.fillna(data_missing_wo_genre.mean()
)
```

```
print("Dataset that has been processed Handling Missing Values with
Mean :")
```

```
print(data_filling.head(10))
```

```
# Observe the missing value imputation in corresponding rows.
```

```
#
```

```
Dataset with empty values! :
```

	CustomerID	Age	Annual Income (k\$)	Spending Score (1-100)
0	1	19.0	15.0	39.0
1	2	NaN	15.0	81.0
2	3	20.0	NaN	6.0
3	4	23.0	16.0	77.0
4	5	31.0	17.0	NaN
5	6	22.0	NaN	76.0
6	7	35.0	18.0	6.0
7	8	23.0	18.0	94.0
8	9	64.0	19.0	NaN
9	10	30.0	19.0	72.0

```
Dataset that has been processed Handling Missing Values with Mean :
```

	CustomerID	Age	Annual Income (k\$)	Spending Score (1-100)
0	1	19.000000	15.000000	39.000000
1	2	38.939698	15.000000	81.000000
2	3	20.000000	61.005051	6.000000
3	4	23.000000	16.000000	77.000000
4	5	31.000000	17.000000	50.489899
5	6	22.000000	61.005051	76.000000
6	7	35.000000	18.000000	6.000000
7	8	23.000000	18.000000	94.000000

8	9	64.000000	19.000000	50.489899
9	10	30.000000	19.000000	72.000000

Filling with Median

The median is used when the data presented has a high outlier. The median was chosen because it is the middle value, which means it is not the result of calculations involving outlier data. In some cases, outlier data is considered disturbing and often considered noisy because it can affect class distribution and interfere with clustering analysis.

```
print(data_missing_wo_genre.median())
print("Dataset with empty values! :")
print(data_missing_wo_genre.head(10))

data_filling2=data_missing_wo_genre.fillna(data_missing_wo_genre.median())
print("Dataset that has been processed Handling Missing Values with Median :")
print(data_filling2.head(10))
```

Observe the missing value imputation in corresponding rows.
#

```
CustomerID      100.5
Age              36.0
Annual Income (k$)  62.0
Spending Score (1-100)  50.0
dtype: float64
```

Dataset with empty values! :

	CustomerID	Age	Annual Income (k\$)	Spending Score (1-100)
0	1	19.0	15.0	39.0
1	2	NaN	15.0	81.0
2	3	20.0	NaN	6.0
3	4	23.0	16.0	77.0
4	5	31.0	17.0	NaN
5	6	22.0	NaN	76.0
6	7	35.0	18.0	6.0
7	8	23.0	18.0	94.0
8	9	64.0	19.0	NaN
9	10	30.0	19.0	72.0

Dataset that has been processed Handling Missing Values with Median :

	CustomerID	Age	Annual Income (k\$)	Spending Score (1-100)
0	1	19.0	15.0	39.0
1	2	36.0	15.0	81.0
2	3	20.0	62.0	6.0
3	4	23.0	16.0	77.0
4	5	31.0	17.0	50.0
5	6	22.0	62.0	76.0
6	7	35.0	18.0	6.0

7	8	23.0	18.0	94.0
8	9	64.0	19.0	50.0
9	10	30.0	19.0	72.0

Part 1: Privacy & Data Lifecycle Analysis of First Assessment

1.1 Data Minimisation Audit

In the pass task 1 I just submit, I made a security device using the sensor PIR motion sensor and accelerometer. When I'm testing my device I captured accelerometer x/y/z and PIR motion 5 times every second.

Before deciding this sample rate out in my program, I have thought about at what rate is the capture is a waste or too much. And I decided to use 5 times per second because in my case my device is use for preventing the user item being touch or has anyone entry the area. This touching or entering activity can be happened very fast, therefore we need a high detect frequency to check if anyone touch user item. However there might be some waste since normally if a person want to enter or touch the item or room, they still need to open the door or enter the area. During these extra step people it cannot touch item that fast. So I think there are still some waste of the detection time.

For sampling frequency, I think using 1 time per sec will be better for long run without losing patterns. For 5Hz = 18,000 row data and 1Hz = 3,600 row data.

I collect for around an hour and I think 10 minutes would have the same insight. As long as someone get detected by my device or move it. however changing from 5Hz to 1Hz would miss some detail like if someone briefly move my device, it might not able to detect. Because 1 sec gap might completely miss the movement.

1.2 Data retention & Lifecycle

I have made a python code to let my data store in local CSV file on my computer. For current stage, only I can access the csv file or people who have my computer password. The csv file is still in my computer. Looking at my csv my timestamp look like “0:54:24”, I didn’t include the real day in there so it hard to re-identify. However there are still a way to find out if the hacker is from the project related staff. Because they know what date the testing is on, so they can use timestamp to find out who is in this area at that time.

1.3 Ethical Stakeholders & Consent

When I am recording the data, me and my family data are captured. I did told them I am doing the project that will capture human movement and they all know. If the csv file misuse, the hacker will know what time we will be at home and what time we are out in the detected area, so they know our living pattern and might break in our house without anyone knowing.

privacy notice

"This area uses motion sensors and accelerometer monitors to secure personal belonging . Data is anonymised (no names, IDs). Raw readings are retained for 7 days, then aggregated to daily averages and retained for 90 days. Room occupants can request data deletion. Facility managers will receive weekly summaries only. Your comfort feedback (optional) is used to refine thresholds.

Part 2: Literature Review & Research Gap Identification

2.1 Literature search

A Low-Cost Vehicle Anti-Theft System Using Obsolete Smartphone

<https://onlinelibrary.wiley.com/doi/full/10.1155/2018/6569826>

Smart Home Anti-Theft System:

<https://www.mdpi.com/2571-5577/1/4/42>

2.2 Synthesis & Gap Analysis

Source	Problem Addressed	Methods Used	Data Capture	Key Finding	Limitation
Liu, Bang, Liu, Nianbo, Chen, Guihai, Dai, Xili, Liu, Ming, <i>A Low-Cost Vehicle Anti-Theft System Using Obsolete Smartphone, Mobile Information Systems</i> , 2018, 6569826, 16 pages, 2018. https://doi.org/10.1155/2018/6569826	High cost of commercial vehicle anti-theft systems and tracking devices	• Velocity-Assisted Dead Reckoning (VA-DR) • Map-matching algorithms • LSTM neural network • Ad-hoc authentication	• Built-in smartphone IMU (accelerometer & gyroscope) • GPS /Wi-Fi signals • Driving history logs	Low-cost tracking achieves localization errors lower than visual sight range	Requires continuous power supply & struggles with prolonged GPS loss without distinct road turns
Pandya, S.; Ghayvat, H.; Kotecha, K.; Awais, M.; Akbarzadeh, S.;	● Inability to detect intruders in total	• IoT framework & smart video analytics.	● Real-time video feeds from standard	● Successfully detects intruders wearing facial	● Dependent on a stable internet/network

Gope, P.; Mukhopadhyay, S.C.; Chen, W. Smart Home Anti-Theft System: A Novel Approach for Near Real-Time Monitoring and Smart Home Security for Wellness Protocol. <i>Appl. Syst. Innov.</i> 2018, <i>1</i>, 42. https://doi.org/10.3390/asi1040042	darkness or when faces are covered (masks/s carves). ● Standard security systems only record incidents after they happen.	• Near real-time motion and human detection algorithms. • Masked-face detection techniques. • Low-light image enhancement processing.	CCTV cameras. ● Physical IoT sensors	coverings. ● Significantly reduces false alarms and sends near instant mobile alerts. ● Achieves reliable intruder detection even in zero-light/pitch-black environments.	connection. ● Processing speed depends on local cloud/hardw are capabilities. ● Video quality under extreme low-light can affect detection accuracy.
--	--	---	---	--	---

2.3 Identify Your Research Gap

Gap 1: False Alarms from Normal Car Movement and No Way to Check Who Entered the Car

The system detects theft simply by feeling the car move through the phone's motion sensors.

Gap: The study does not combine this motion tracking with other simple checks like detecting if a car door was opened, if a window was smashed or if an authorized driver's phone/key is nearby. Because of this, normal everyday events like a tow truck moving the car, strong wind, heavy traffic vibrations or a family member borrowing the car can trigger false theft alerts. Without a way to double check if an actual break in happened inside the car, the system causes unnecessary panic.

Gap 2: Not Tested in Real-World Weather or multiple Lighting

The authors tested theft detection under controlled low light and face cover conditions. However, they didn't test how the system handles real-world disruptions like passing car headlights, flickering lights, dark shadows, heavy rain, fog, bugs crawling on the camera, or curtains blowing in the wind.

Gap: Right now, the system is only tested in quiet indoor rooms. Its ability to spot shapes already drops from 97% in normal light down to 66% in total darkness. Will the camera break down or keep failing if it has to deal with outdoor weather, changing light, or random moving objects like tree branches?

Gap 2: Needs Internet to Work and Has No Backup Plan If Offline

The system depends heavily on sending live video through the internet to a central cloud server to send alerts to the owner's phone.

Gap: The study never tests what happens if the internet is slow, weak, or cut off—or if a burglar uses a signal jammer. If the internet or power goes out, the system cannot process the video or save backup footage locally, meaning it completely fails to send alerts when you need them most.

Part 3: Enhanced Use Case Design with Hypotheses

3.1 Refined Problem Statement

Original Problem Statement:

How can we build an affordable anti-theft device using a PIR sensor and accelerometer to catch someone touching or moving personal belongings on a desk by triggering local lights and sound?

Refined Problem Statement:

Simple security devices often make mistakes and trigger **false alarms**. For example, the buzzer might go off just because someone walked past the desk or gently bumped the table. When false alarms happen all the time, people get annoyed and turn off the device.

Main Research Question:

How can we combine data from two sensors, the PIR motion sensor (which sees body heat) and the LSM6DS3 accelerometer (which feels physical shaking) so the device can tell the difference between normal room activity and actual theft while reducing false alarms by 40%?

3.2 Sensor & Data Capture Protocol Design

1. Hardware Configuration

- **PIR Sensor (Spatial Proximity):** Detects body heat and human movement near the monitored desk or personal area.
- **LSM6DS3 Accelerometer (Physical Kinematics):** Built into the Arduino Nano 33 IoT board to detect 3-axis physical motion (X, Y, Z axes), tilting or direct snatching of personal belongings.

2. Enhanced Data Capture Specifications

Sampling Rate (Reading Speed): 1 per second

Sampling 1 time per second reduces data volume to **3,600 rows per hour**.

Physical theft or laptop movement lasts several seconds, so 1 Hz clearly captures the movement pattern without filling up board memory or crashing the system.

Collection Period: Data is collected 1 day (comparing normal room activity against simulated theft tests) to evaluate performance during both quiet nights and busy daytime hours.

Storage: local CSV

Data fields:

Each data row records the following specific fields:

1. Timestamp: Time elapsed in milliseconds (ms).
2. PIR_State: Motion status (0 = No body heat motion, 1 = Motion detected).
3. Accel_X, Accel_Y, Accel_Z: Raw acceleration along the 3 axes.
4. Accel_Delta: Total movement change magnitude calculated to measure shaking/movement intensity.

Privacy:

No Biometric or Personal Data: The device only logs simple numbers (infrared heat states and 3-axis movement). No cameras, microphones or user names are used.

Data Cleanup: Raw log files are used strictly for testing the detection algorithm and deleted after the project is finished.

3.3 Hypotheses

H1: Using Two Sensors Together is Better

Statement:

Using both sensors at the same time (PIR motion + LSM6DS3 accelerometer) will

catch real theft over 95% of the time and cause very few false alarms (less than 5%) compared to using only one sensor.

Simple Explanation:

When used alone, the PIR sensor triggers false alarms when people just walk past. The accelerometer triggers false alarms when someone bumps the table. Checking both sensors together stops these mistakes.

How to Test It:

Try 50 test runs (25 fake theft attempts and 25 normal daily activities like walking past or working at the desk). Count how many false alarms happen when using one sensor vs. both sensors together.

H2: Short Bumps vs. Real Theft

Statement:

Small accidents (like typing or lightly bumping the desk) only shake the sensor for a fraction of a second (< 1 second), but real theft (grabbing and moving an item) moves the sensor for at least 2 or 3 seconds in a row.

Simple Explanation:

A quick desk bump happens too fast to trigger an alarm at 1 Hz reading speed. Stealing an item takes longer, so the sensor will see movement across multiple seconds in a row.

How to Test It:

Compare the saved data logs of quick table bumps against real laptop movement to check how many seconds the shaking lasts.

3.4 Theoretical Validation

A. Literature-Based Validation

Alignment with Existing Theory:

Our hypotheses align directly with established Multi-Sensor Data Fusion (MSDF) frameworks and physical security literature.

Literature Support:

PIR Limitations: Research shows single PIR sensors suffer from high false-positive rates due to convection currents, HVAC airflow, or direct sunlight.

Accelerometers: Motion science shows that when someone moves or steals an object, the physical movement changes the total movement numbers

Application:

Checking heat motion (PIR) and physical shaking (accelerometer) together, checking how long the movement lasts, and using a total movement math formula all follow proven science to stop false alarms.

Statistical Feasibility

Sample Size Support:

Collecting data at 1 Hz yields 3,600 readings per hour, which equals 86,400 readings per day

Oversampling vs. Statistical Needs:

Standard statistical tests (such as paired t-tests or logistic regression) only require $n \geq 30$. Our dataset has massive oversampling, providing more than enough statistical power to test detection accuracy ($> 95\%$) and false alarm rates ($< 5\%$).

Statistical Assumptions to Test:

Temporal Autocorrelation: Consecutive 1 Hz readings are correlated over time. We will check time-series dependence and apply rolling moving averages.

Normality & Linearity: Baseline idle sensor noise will be tested for normal distribution to set accurate trigger thresholds.

C. Sensor Validation

Both sensors are ideal for this task.

LSM6DS3 Accelerometer Fitness:

The sensor datasheet shows it measures up to $\pm 16g$ of movement with low noise. Stealing an item creates movements over $0.3g$, so the sensor is plenty sensitive to catch a thief.

PIR Sensor Fitness:

The PIR sensor sees movement up to 7 meters away.

The PIR datasheet states it needs 10 to 30 seconds to warm up when turned on. the code will automatically throw away the first 30 seconds (30 readings) after bootup so it does not trigger a fake start-up alarm.

D. Data Quality Checks (Planned)

If the computer disconnects briefly, missing data under 5% will be filled using the last known good reading (ffill). If more than 5% of data is missing in an hour, that mean entire hour of data will be thrown out.

Outlier Strategy is Any acceleration reading higher than 16g which is impossible for normal human desk movements or negative time values will be flagged as sensor errors. These bad readings (less than 1% of total data) will be deleted and replaced with average values.

3.5 Monitoring & Real-Time Notification Design

To elevate our security device from a simple local alarm into a smart live security system, I will design a real-time monitoring dashboard and alerting engine.

My Plan:

Platform	Arduino Cloud Dashboard
Widgets	Live PIR state (Occupancy), Acceleration Magnitude (Movement), Security Status (Safe/Alert)
Warning Alert	PIR motion + Item movement ($\geq 2s$) $\rightarrow$ LED + Email/Browser push notification
User Privacy	Clear desk notification sign + Zero camera/audio logging

3.6 Revised Data Lifecycle & Ethics

Data Minimisation

In my first testing in P1, I recorded sensor data **5 times every second (5 Hz)**. This created 18,000 rows of data every hour. I realized that collecting data this fast creates too much wasted data. In this updated design, I reduced the reading speed to **1 time per second (1 Hz)**. This cuts the data down to 3,600 rows per hour, saving computer memory while still catching multi-second theft movements. I also strictly limit the data collected to simple numbers: Timestamp, PIR_State, raw 3-axis acceleration (Accel_X, Accel_Y, Accel_Z), and Accel_Delta. The system collects zero personal names, student IDs, camera video, or sound recordings.

Resubmit Answer

1. How does the system tell the difference between the Owner and an Intruder?

Since the system does not use cameras or face scanners, it tells the owner and intruder apart using simple rules and timing.

Manual Turn-Off: The owner can turn off the alarm before touching their item (for example, by pressing a hidden switch, using a phone app or flipping a switch on the online dashboard). If the item moves while the alarm is still turned ON, the system knows it is an intruder.

This is the most simple and cost effective way to separate a theft and a owner.

2. How can PIR and Accelerometer replace Video/Cameras?

The research paper uses CCTV cameras and video software to spot intruders. system replaces expensive cameras by measuring **heat and physical movement** instead of taking pictures.

PIR can detect near human movement and accelerometer can check if the object being move or not. using these two sensor, they can replace difficult cameras detection which require complex code to track human activity and limitation of object movement. And camera can be affect by different lighting as well.

Data Retention Policy

Raw Data (0–7 Days): Raw 1 Hz CSV log files are stored on my password-protected computer for 7 days to test the detection program and verify that false alarms are reduced.

Daily Summaries (8–90 Days): After 7 days, all raw sub-second files are converted into daily average summaries (such as total trigger counts and peak movement levels). The detailed raw 1 Hz files are permanently deleted.

Final Erasure (After 90 Days): All remaining summary files are permanently erased from my computer after 90 days.

Occupant Consent

Because testing happens in home spaces with family members around, informed consent is very important. Unprotected motion logs could reveal daily family routines, showing hackers when our home is empty. To prevent privacy issues, the following clear sign will be placed next to the device:

PRIVACY NOTICE

This area uses motion sensors and accelerometer monitors to secure personal belongings. Data is anonymised (no names, IDs). Raw readings are retained for 7 days, then aggregated to daily averages and retained for 90 days. Room occupants can request data deletion. Facility managers will receive weekly summaries only. Your comfort feedback (optional) is used to refine thresholds.

Stakeholders & Power Dynamics

Who Benefits: Item owners gain security for their belongings. Room occupants benefit because combining two sensors reduces annoying false alarm noise from the buzzer.

Power Imbalances & Fixes: A power imbalance happens if housemates or room occupants feel secretly watched or tracked. We fix this by using zero cameras or microphones and placing a clear privacy sign near the device. Occupants can also ask to delete their test data at any time.

"This assessment work was developed with assistance from the following approved Deakin genAI tools:

- Google Gemini (accessed July 2026): Used to ask research and summarize content for easier understanding.

Used to All genAI-generated suggestions were critically evaluated and integrated only where supported by evidence. The final design, decisions, and justifications are my own."

AI Ginimi output:

<https://onlinelibrary.wiley.com/doi/full/10.1155/2018/6569826> summary:

This link leads to the open-access research paper "**A Low-Cost Vehicle Anti-Theft System Using Obsolete Smartphone**" published in *Mobile Information Systems* (Wiley/Hindawi, 2018) by Bang Liu, Nianbo Liu, Guihai Chen, Xili Dai, and Ming Liu.

Core Concept: *PhoneInside*

Instead of buying expensive, specialized vehicle tracking hardware, the authors propose **PhoneInside** — a framework that repurposes an unused, old smartphone into an automated anti-theft and tracking unit.

Key Technical Features

- **Repurposed Hardware Setup:** An obsolete Android smartphone is mounted inside the car's body and connected to a car charger for continuous power.
- **Theft Detection & Alerts:** Uses internal sensors (accelerometer, gyroscope) alongside an ad-hoc authentication protocol to detect unauthorized vehicle movement. If theft is detected, it immediately sends real-time alert notifications and location updates to the owner.
- **Adaptive Multi-Source Localization:** Combines GPS, cellular/Wi-Fi positioning, and inertial sensors to ensure continuous tracking, even when satellite signals are blocked (e.g., in parking garages or tunnels).
- **Novel VA-DR (Velocity-Assisted Dead Reckoning):** Standard dead

reckoning accumulates positional drift over time. The paper introduces a VA-DR algorithm that uses map data and turn angles at intersections/curves to estimate velocity and correct trajectory errors dynamically.

Main Contributions & Results

1. **E-Waste Reuse:** Demonstrates a practical, low-cost way to reduce electronic waste while providing property protection.
2. **High Tracking Accuracy:** Experimental testing showed that the system tracks stolen vehicles accurately, keeping average localization errors well within visual line-of-sight range.
3. **Cost-Effective:** Achieves performance comparable to commercial OBD-II trackers or dedicated GPS units at a fraction of the cost.

1. Map-Matching Algorithms & Dead Reckoning

- **Velocity-Assisted Dead Reckoning (VA-DR):** Calculates trajectory without an OBD-II connection by combining inertial sensor data with street layout geometries.
- **Topological Map-Matching:** Aligns raw GPS/location points onto known digital road segments to eliminate drift.

2. Signal Processing & Classical State Estimation

- **Sensor Filtering:** Preprocesses raw accelerometer and gyroscope data to remove noise and gravity biases before calculating vehicle turns or movement.
- **Finite State Machine (FSM):** Manages system transitions between monitoring modes (e.g., *Parked*, *Vibration Detected*, *Theft Warning*).

3. Machine Learning (Deep Learning)

- **Long Short-Term Memory (LSTM) Network:** Learns the owner's driving history to build a custom "mobility model." It analyzes driving patterns to detect abnormal driving behavior characteristic of theft.

<https://www.mdpi.com/2571-5577/1/4/42> give me a easy summary what it about

This paper is titled "**Smart Home Anti-Theft System: A Novel Approach for Near Real-Time Monitoring and Smart Home Security for Wellness Protocol**".

What It's About in Simple Terms

1. **The Core Idea:** It proposes an **Internet of Things (IoT)-based smart home security system** designed to catch burglars in real time and immediately notify the homeowner on their device so action can be taken right away.
2. **The Problem with Standard Security Systems:**
 - Traditional CCTV systems usually just record video footage, meaning you only discover a theft **after** it has already happened.
 - Old systems struggle to tell the difference between humans and pets/objects, frequently trigger false alarms, or fail if an intruder covers their face with a mask, hat, or scarf.
3. **How This System Fixes It:**
 - **Detects Hidden Faces:** Uses smart detection algorithms to spot human intruders even if their face is partially or fully covered.
 - **Works in the Dark:** Can detect intruders in pitch-black conditions using standard CCTV cameras without needing expensive night-vision gear.
 - **Instant Alerts:** Combines physical sensors with cloud software to process video data quickly and send near-instant alerts to the homeowner's phone.

i am confuse, explain what i need to do in P2 Part3 in easy way.

What is Part 3 asking you to do?

Think of Part 3 as taking your simple P1 anti-theft security device (the PIR sensor + LSM6DS3 accelerometer) and designing a real, scientific test for it.

Instead of just saying *"I built a device that beeps when touched,"* Part 3 asks you to explain:

1. What **specific problem** you are trying to solve.
2. How you will **collect sensor data** properly.
3. What **educated guesses (hypotheses)** you are testing.
4. Why your plan makes sense **theoretically and ethically**.

Step-by-Step Breakdown (Made Easy)

1. Refined Problem Statement

- **What to do:** Take the problem from P1 and make it more specific.
- **Simple Idea:** "PIR sensors often trigger false alarms when people just walk past or when the heater turns on. How can we combine the PIR sensor and the LSM6DS3 accelerometer to catch actual theft while reducing false alarms by 40%?"

2. Sensor & Data Capture Protocol Design

- **What to do:** Fill out or write down the exact rules for how your device gathers data.
- **Simple Idea:**
 - **Which sensors?** PIR (detects body heat/proximity), LSM6DS3 (detects physical moving/shaking), and optional DHT22 (detects room temperature changes).
 - **How fast?** Sample PIR and IMU at **10 Hz** (10 times per second).
 - **For how long?** Collect data over 14 days (or simulated test runs).

- **What data?** Time, $X/Y/Z$ acceleration values, motion state (HIGH/LOW), and room ID.
- **Privacy:** Don't record personal names, camera video, or exact GPS locations.

3. Hypotheses (Your Educated Guesses)

- **What to do:** Write 2–3 clear statements that you can test with numbers (H_1, H_2, H_3).
- **Simple Idea:**
 - H_1 (**Main Guess**): Combining PIR motion *AND* accelerometer movement at the same time will be much more accurate ($> 95\%$) than using just one sensor alone.
 - H_2 (**Vibration Guess**): Someone bumping the desk makes a quick spike (< 200 ms), but someone stealing the laptop makes a continuous movement (> 600 ms). A moving average filter will ignore the quick bump.
 - H_3 (**Temperature Guess**): Sudden changes in air conditioning cause false PIR triggers when no one is in the room.

4. Theoretical Validation (Pre-Execution Checks)

- **What to do:** Prove that your setup makes sense *before* running the full test.
- **Simple Idea:**
 - Show the math formula you use to measure movement regardless of how the device is sitting on the desk:

$$|\vec{A}| = \sqrt{A_x^2 + A_y^2 + A_z^2}$$

- Explain why you clean the data (e.g., throw away corrupt sensor readings or apply a moving average filter to reduce noise).

5. Monitoring & Real-Time Notification Design

- **What to do:** Draw or explain what happens when motion is detected.
- **Simple Idea:**
 - **Level 1 (Slight Motion):** Someone walks near → Blink a low LED warning.
 - **Level 2 (Theft / Heavy Movement):** Someone picks up the laptop → Sound the buzzer + send a push notification to your phone via Arduino IoT Cloud.

6. Revised Data Lifecycle & Ethics

- **What to do:** Explain how you keep people's data safe and private.
- **Simple Idea:**
 - Delete raw high-speed sensor data after 7 days.
 - Create a simple **Privacy Notice** sign to stick near the desk so room occupants know only motion and acceleration are logged (no audio or video).

Quick Checklist to Complete Part 3

1. **Define your problem clearly** (Focus on stopping false alarms using sensor fusion).
2. **Specify your data settings** (10 Hz sampling rate, 14 days, privacy protection).
3. **Write 2 to 3 testable hypotheses** (H_1, H_2, H_3).
4. **Explain your math/filters** (Moving average filter & magnitude formula).
5. **Describe alert levels** (Local light/sound vs. Remote phone alerts).
6. **Add an ethics notice** (Inform users what data is collected).